# CTSpinoPelvic1K: spine, pelvis, ribs and femora in one coordinate frame, annotated for lumbosacral transitional anatomy

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**Purpose:** A vertebra at the lumbosacral junction is named by counting. The gold standard counts caudally from C2 on whole-spine imaging[1]; but in practice, a lumbar surgical case is planned on lumbar-only imaging[2–4] (T12 to S1) without C2. Abdominopelvic computed tomography (CT) holds that span plus the lowest ribs and pelvis, making it a reasonable stand-in for the preoperative view. When a lumbosacral transitional vertebra (LSTV) alters the count, the local anatomy is ambiguous. Four rib-free vertebrae may be an L1 with a lumbar rib or an L5 assimilated to the sacrum, and six may be a sixth lumbar vertebra, a T12 with aplastic ribs, or a lumbarized S1[5]. CTSpinoPelvic1K asks whether local morphology can resolve that ambiguity without the count. Two public label sets already covered this imaging, CTSpine1K's vertebrae and CTPelvic1K's pelvis, on the same colonography patients, never joined; this release joins them on one series and annotates the bones neither had. It provides 802 CT records, per-level ribs and femora, lumbar levels anchored on the lowest rib-bearing vertebra and the sacrum, and classes for L6, T13 and lumbar ribs, so anomalies are recorded as themselves, not forced into whichever normal count they happen to resemble.

**Acquisition and Validation Methods:** Records pair CTSpine1K[7] and CTPelvic1K[8] labels on each patient's bone-richest acquisition under a VerSe-native scheme. Validation covered geometric invariants (802/802 pass), rib–vertebra incidence across 5,749 ribs (0.035% offset), and spinopelvic measures matching published values.

**Data Format and Usage Notes:** NIfTI image/label pairs with patient-grouped LSTV-stratified five-fold splits and a loader; archived at https://doi.org/10.5281/zenodo.22139642.

**Potential Applications:** Classifying a vertebra from local features; updating cadaveric morphometry; spinopelvic assessment; opportunistic screening; and, absent a public preoperative lumbar cohort, surgical planning research (377 records prone). *Limitations:* thoracic ground truth is field-of-view limited; postural angles are supine; no held-out test set; ribs are triaged-review pseudolabels; Castellvi grades are two-reader consensus on 33 records.

## I. INTRODUCTION

The spine commonly carries seven cervical, twelve thoracic and five lumbar vertebrae, and a vertebra is identified by where it falls in that sequence. Transitional variants sit at the two ends of the lumbar column, the thoracolumbar transitional vertebra (TLTV) above and the lumbosacral transitional vertebra (LSTV) below. Both disturb identification in two ways at once. First, they change what the border vertebra and its ribs *look like*: a thoracolumbar border vertebra may carry a full rib, a hypoplastic stump, or none at all, and a lumbosacral one may be partly or completely assimilated to the sacrum (*sacralization*), with an anteriorly wedged body, a reduced disc beneath it and hypoplastic facet joints, or, conversely, separated from it (*lumbarization*), with a squared body, lumbar-type facets and a full-height disc[5]. Second, they change *how many* vertebrae each region contains, eleven or thirteen thoracic and four or six lumbar.

Either change alone would be tractable, but together they make it difficult to state definitively what any vertebra at these borders *is*. Each of those readings actually yields the same count (Fig. 2), and the morphology that would separate them is not decisive to a human reader in every case. Whether it differs enough for a learned classifier to separate them, or at least to assign each reading a probability, is the question this release is built to support. Short of that, what remains is the count itself. The gold standard for numeration in transitional anatomy is whole-spine imaging, counting caudally from C2[1], which fixes how many vertebrae and how many rib pairs the column holds, and no spine-limited acquisition can supply either count. LSTV is common, reported at 10–29% in the general population[6] and in 16.3% of a whole-spine CT series,[9] and wrong-level surgery is its most serious consequence.

Existing public collections cannot address this, and not for lack of size (Table I): spine datasets omit the pelvis, pelvic datasets do not number the vertebrae, TotalSegmentator has no class for a sixth lumbar vertebra or for a rib on a lumbar vertebra, and VerSe, the one collection with an L6 class, stops at the sacrum and carries no ribs. None of these collections, then, can record a six-lumbar spine together with both of its borders, and a segmenter trained on them must, when it encounters an enumeration anomaly, shift the whole column by a level or absorb a vertebra into its neighbor. The size of that problem has now been measured on CT, where prior labeling methods assigned every vertebra correctly in 77% of subjects and an anomaly-aware extension of SPINEPS[10] raised that to 99%[11]. Its weights label L6 and T13 on CT, but its 1,536-scan CT cohort is in-house and unreleased, so what exists is a capability, not a dataset.

The intraoperative side has prior art of its own. LevelCheck registers the intraoperative radiograph to the preoperative CT and projects the CT's vertebral labels onto it[12–14], but those labels are placed by hand and verified by the surgeon[12], so the method assumes a correctly labeled CT and does not supply one. Automating that step for anatomy that itself is anomalous is what a dataset carrying the anomaly classes is for.

CTSpinoPelvic1K places the spine, pelvis, ribs and femora in one coordinate frame and gives a class to each structure an enumeration anomaly produces, a sixth lumbar vertebra and a rib borne by a lumbar vertebra. A scheme without those classes simply lacks a class vocabulary expressive enough to describe anatomic anomalies.

The project began with a surgeon's question, whether a vertebra can be named from the anatomy in front of the reader, on the images a lumbar surgical case is actually planned on. By guideline those are lumbar-only[2–4,15]; whole-spine coverage is reserved for trauma with an identified injury[16,17] and for deformity radiographs. An abdominopelvic CT runs diaphragm to pelvic floor, so it captures that span with the lowest ribs and the whole pelvis, and lacks C2 exactly as the planning study does. Every record comes from one prospective trial protocol, ACRIN 6664[18,19], which scanned 802 patients aged 50 and over, supine and prone, on multidetector CT at 15 centers, on five scanner vendors and nine models, with manufacturer, model and reconstruction kernel recorded per record. To our knowledge, it is the largest spine-and-pelvis annotated CT cohort acquired under a single protocol; the comparators, by contrast, pool collections, are multi-site by design, or are routine clinical scans under many protocols (Table I). With the protocol held fixed, scanner effects on a model become measurable rather than confounded. The price is an abdominal field of view, in which only the lowest thoracic levels are present (Sec. V).

The gap was evident. CTPelvic1K[8] and CTSpine1K[7], released months apart by an overlapping author group, each annotated the COLONOG collection under radiologist supervision, one the pelvis and the other the vertebrae, on the same patients, yet no one joined them, although the join needs no new imaging and no new radiologist. It is not a file merge, because each annotation had to be traced back to the series it was drawn on (Sec. II.A). Once completed, that crosswalk yielded a spine-and-pelvis frame at 802 patients, and the bones neither set had, ribs, femora and hardware, could then be annotated against it rather than from scratch.

### I.A. Vertebra labeling when the scan cannot settle the count

The main limitation is that no scan in this corpus contains C2, so the conventional count cannot be performed, a property of abdominal imaging rather than of the annotation. A thirteenth thoracic vertebra and a lumbar rib are different phenotypes, distinguished as separate subtypes in cadaveric classifications of the thoracolumbar junction[20,21]: the first is an additional rib-bearing segment, usually above five lumbar vertebrae, so the column holds 25 presacral vertebrae; the second is a rudimentary rib on a lumbar-type L1 in a column of the usual 24. Neither is separable by counting at the two borders used here, the last rib-bearing vertebra and the sacrum, because four rib-free vertebrae can fit a lumbar rib, a cranially shifted T13 or an L5 assimilated to the sacrum, and six can fit an aplastic twelfth rib or a sixth lumbar vertebra. A *stump rib*, hypoplastic and a cardinal indicator of a thoracolumbar transitional vertebra, is still a rib, so only its length records the anomaly. Rib anomalies travel with the lumbosacral border: on whole-spine CT hypoplastic twelfth ribs occur with sacralization and lumbar ribs with lumbarization.[9]

In this release, every vertebra carries the identifier its radiologist-sourced annotation assigned, corrected where the source was wrong; those labels are the ground truth. In addition, every released measurement is expressed against the two landmarks present in essentially every abdominal scan, the sacrum below and the lowest rib-bearing vertebra above, so that the measurement is positional rather than nominal: a vertebra by how many rib-free vertebrae separate it from the sacrum, a lowest rib by its length as a fraction of the rib above it, a transverse process by its span and its distance from the ala. None of these values shift with the name a reader gives the junction, so cases that disagree about the name remain comparable.

*But the vertebrae themselves are known to differ.* Level-specific morphometry is long established: pedicle width and height change systematically from the tho-

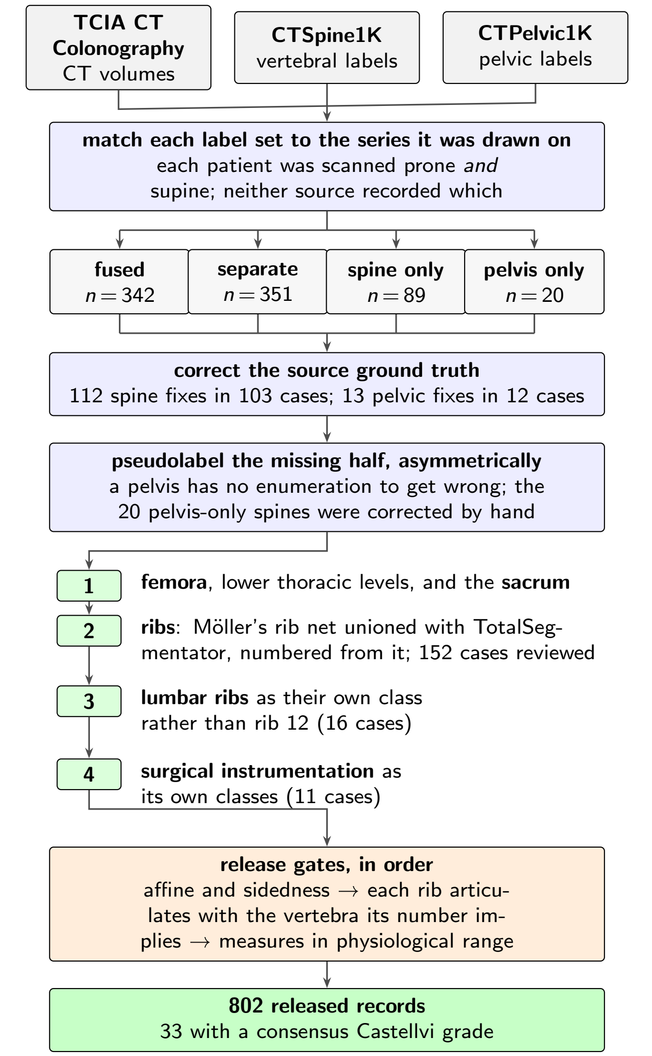


FIG. 1 How the dataset was built. $n$ counts scans, or corrections, at each step.

racic to the lumbar spine,[22] vertebral body, endplate and canal dimensions differ by level,[26–28] and the transverse process and the iliolumbar ligament mark the last lumbar vertebra independently of any count.[5,29] If those differences hold at the boundary itself the phenotype is decidable locally, and on this corpus it is. Separating T12 from L1 on shape alone gives an area under the curve of 0.990 and 97.3% accuracy over 1485 vertebrae: a logistic regression on five released per-level measures and six ratios between them, each measure divided by the same patient's median across their levels, so that size, which separates thoracic from lumbar trivially and says nothing at the junction, is removed; folds are grouped by case and the curve is taken on pooled out-of-fold scores (`test_morphometric_separability.py` in the released code). Schinz et al. reach the same conclusion from the other side: on 1,242 whole-thoracolumbar CTs a shape-based labeling of the junction matched nerve morphology in every case, against 92.6–97.2% for counting- and rib-based rules.[30]

## II. ACQUISITION AND VALIDATION METHODS

### II.A. Sources, and why a crosswalk was necessary

Both sources draw part of their imaging from The Cancer Imaging Archive (TCIA) CT colonography collection (COLONOG)[18,19]. CTSpine1K's 1,005 volumes come from four collections and CTPelvic1K's 1,184 from seven; COLONOG is the only collection common to both and the only one acquired under a single protocol, meaning one scan prescription and patient preparation for every case, so this release is its COLONOG subset, 782 patients with a CTSpine1K vertebral annotation and 20 with a CTPelvic1K pelvic annotation only, 802 in all. Neither source released the mapping from an annotation to the specific CT series it was drawn on.

That mapping matters because every patient in CT COLONOGRAPHY was scanned *twice*, prone and supine, and an annotation carries a patient identifier and nothing finer, so neither collection recorded which acquisition it had labeled. Turning a patient from supine to prone alters lumbar lordosis and the alignment of each segment against the next, so outlines drawn on one series are the wrong *shape* for the other and no rigid realignment recovers them. The crosswalk (Fig. 1) therefore resolves each annotation to a patient and then, within that patient's volumes, to the series whose bone *agrees* with it, with candidates thresholded at bone attenuation and scored by overlap with the mask. This step mattered for 351 patients, nearly half the cohort, whose two collections had labeled different acquisitions of the same person; those records are released but held out of any analysis assuming a single posture, since two labeled postures of one patient are a resource rather than a defect.

### II.B. Densification, and why it is asymmetric

A corpus in which half the records lack a pelvis is not a spinopelvic dataset, so the missing structures were pseudolabeled, asymmetrically (Fig. 1). Pelvis onto spine-only records is the safe direction, since the sacrum and innominate bones carry no enumeration to get wrong, and quality is reported as held-out performance on the *pelvic_only* records, withheld from training for that purpose. The reverse direction carries the whole problem this dataset addresses, because a pseudolabeled spine must commit to a count, and on a transitional vertebra it commits to the commoner reading; those 20 records were corrected by hand.

### II.C. Ribs

No public rib annotation covers this cohort, and the two available automatic sources[34,35] fail in complementary ways.

Möller's binary rib network[35] reports high accuracy, and its authors show that stump ribs can be classified from a partial field of view; that result concerns the morphological features computed on a rib once segmented.

Table I Public CT collections at the lumbosacral junction. Black, class present; outlined, graded only to exclude it[31].

| Collection | Scans | Count | L6 | Sacrum | S1 | Pelvis | Ribs | L. rib | Femora | Grade |
|---|---|---|---|---|---|---|---|---|---|---|
| CTSpine1K[7] | 1,005 | ■ | ■ | | | | | | | |
| CTPelvic1K[8] | 1,184 | | | ■ | | ■ | | | | |
| VerSe[32] | 374 | ■ | ■ | | | | | | | □ |
| RibSeg v2[33] | 660 | | | | | | ■ | | | |
| TotalSegmentator[34] | 1,204 | ■ | | ■ | ■ | ■ | ■ | | ■ | |
| **CTSpinoPelvic1K** | 802 | ■ | ■ | ■ | | ■ | ■ | ■ | ■ | ■ |

Here, though, the segmentation itself failed on ribs that were *partially outside* the field of view. A rib crossing the boundary of an abdominal acquisition was liable to be missed altogether rather than segmented to the edge. In a colonography cohort the lower ribs are routinely clipped, and those happen to be the ribs the count depends on.

TotalSegmentator[34], by contrast, does segment clipped ribs and supplies the per-rib numbering that a binary network cannot. Its characteristic failure is at the other end of the bone, where the most medial portion of the rib (roughly the last tenth to sixth approaching the costovertebral joint) is frequently absent. A rib truncated at its medial end never reaches the vertebra it articulates with, so the rib–vertebra incidence check (the quality-control gate on the rib class labels, which matches each rib to the vertebra its number implies) has nothing to test.

Given these complementary gaps, the two approaches were combined (Fig. 1), TotalSegmentator supplying the numbering and the medial-clipped cases, Möller the detection where its output is more complete.

The result is a pseudolabel whose review was triaged rather than exhaustive. A label-based rule flagged every record in which one rib bone carried more than one identifier, or in which a rib failed to reach the vertebra it articulates with, since a single bone carrying two numbers is a numbering error whatever the anatomy. The rule selected 152 records; 148 were reviewed and corrected, and four were not and are identified in the release. Reviewers were medical students working through OpenSpineConsortium[36], trained against a written labeling protocol, with every correction attributed to its annotator. Records passing the rule were not individually inspected; the residual rib–vertebra offsets reported in Sec. II.F are the independent check on what the rule missed.

### II.D. Label scheme and counting anchors

The scheme is VerSe-native. Vertebrae keep their VerSe identifiers (C1–C7 = 1–7, T1–T12 = 8–19, L1–L6 = 20–25, sacrum 26, coccyx 27, T13 = 28), and every non-VerSe structure takes a fixed identifier above that range, hips (30–31), femora (32–33), thirteen ribs per side (34–46 left, 47–59 right), lumbar ribs (60–61) and surgical hardware (62–68). Identifier 29 is retired and unused, for the reason given below. Rib 13 is the true rib of a T13 and is empty in this release, so a thirteenth thoracic vertebra and a lumbar rib are recorded as the distinct phenotypes they are.

Two anchors give every record the same positional frame, and Fig. 2 shows both, the lowest rib-bearing vertebra above and the **sacrum** below. The sacrum's superior end-plate is the landmark sacral slope and pelvic incidence are measured from, and with both brackets present L5 and L6 are distinguishable where a spine-limited view alone would not be.

*A separate S1 label was withdrawn, and why.* Earlier releases carved the cranial portion of the sacrum as its own S1 class, on an automatic estimate of the S1–S2 boundary. Measured against the anatomy on all 802 records, that estimate is unreliable: S1's craniocaudal extent is a median 57.6 mm where a first sacral segment is near 30, only 26% of records fall in a defensible range, and in 222 the plane takes more than half the sacrum. The failures are not merely low cuts. Where the fit degenerates the plane comes out nearly parallel to the sacral long axis and shaves the ventral cortex, so the retained sacrum is left without its anterior wall — wrong for every use of the label, not only for the end-plate. Rendering one such record is what showed this; no summary statistic did.

The consequence was confined to the parameters fitted to that surface. Splitting the cohort on whether the carve was anatomically sound separates them cleanly: the 580 sound records give a pelvic incidence of 52.7° and a pelvic tilt of 19.6°, against Hasegawa's supine-CT 53.4 and 19.2, while the 221 degenerate ones give 68.5 and 33.6, and the end-plate gate described in Sec. II.E fires on 96% of the second group and 4% of the first. The measurement was never at fault; the surface it was given was. So S1 is merged back into the sacrum, restoring the bone the source annotation actually delineated, and the end-plate is fitted to the whole sacrum instead.

VerSe[32] carries L6, whose identifier this release keeps, so its L6 *is* VerSe's.

**A rib on a lumbar body receives its own class**, and this is the one class here with no published counterpart. A lumbar rib and a stump rib are the same object under two counts, so a scheme that numbers every rib 1–12 leaves the annotator with two bad options: either calling it rib 12, which asserts that the vertebra beneath it is thoracic, the very question at issue, or discarding it. TotalSegmentator carries ribs 1–12 per side and no such class, and VerSe's T13 is a vertebra rather than a rib. This scheme records both: VerSe's T13 (28) with its rib pair (46, 59) for a thoracic-type thirteenth vertebra, and 60/61 for a lumbar-type body bearing a rudimentary rib. No record here carries a T13; the sixteen lumbar-rib cases were read as lumbar-type bodies, and where a border vertebra could not be settled by its readers the label stands as read.

Figure 2 shows why the interval is the primitive and the label is derived. Among the records with four rib-free vertebrae, some reach that count from above, through a rib on L1, and some from below, through a lumbosacral segment fused to the sacrum, in Fig. 2(c) together with stump twelfth ribs; the count alone cannot tell them apart, and the release can because ribs, rib lengths and the lumbosacral junction are all labeled.

Sixteen records carry a lumbar rib, thirteen bilateral and three unilateral, all on the right, too few to speak to side preference.

Eighteen records carry an L6. Seventeen of them carry a consensus Castellvi grade; the eighteenth was not graded. The source transitional label (pelvic where CTPelvic1K flagged one, otherwise the lumbar count) reads lumbarization in fourteen, sacralization in two and semi-sacralization in one, and is unremarkable in the ungraded record. A spine can carry six lumbar-type vertebrae and still be read as a sacralization, depending on where the reader started the count.

Both counts, and the manifest fields that report them (`has_l6`, `has_lumbar_rib`, `n_lumbar_labels`, `lumbar_rib_side`), are computed from the released label volumes by counting identifiers 25 and 60–61.

### II.E. Computational tools

*Access.* Every step is performed by open-source code archived with the dataset, together with the reference loader, the label scheme and the quality-control scripts that generated the tables in this manuscript.

*Versions and parameters.* Segmentation of femora, the sacral sub-division and the per-level rib numbering used TotalSegmentator[34] (`total` task, release 2.x) on a single graphics processing unit. The binary rib network is Möller's[35] released weights, applied through the nnU-Net v2 framework[37]. The pelvic completion for records whose pelvis was absent used a five-fold nnU-Net v2 ensemble applied out-of-fold, so that no record is completed by a model that has seen it. The five fold checkpoints, with their nnU-Net plans and dataset descriptor, are released alongside the data (`https://huggingface.co/OpenSpineConsortium/spinopelvic-seg-checkpoints`, trained with `https://github.com/Gregory-Schwing-MD-PhD/spinesurg-ct-nnunet` at commit 50b209f). Image handling throughout used NiBabel and SciPy; no image is resampled or reoriented when a label is written, so every label shares its image's grid and affine exactly.

*Measurement.* Dimensions and spinopelvic parameters come from code released with the dataset. A sacral plate whose fitted normal lies more than 60° off the cranial axis is rejected and the case reported missing, since such a fit has found the near-vertical anterior face of the promontory, not the endplate. Gating on the geometry rather than on the value leaves unusual patients in. OpenSpineToolkit, a unit-tested open-source package is released separately (`https://github.com/OpenSpineConsortium/OpenSpineToolkit`) as a reusable implementation measuring in each vertebra's own frame; no open-source implementation computing these parameters from CT segmentations was found elsewhere.

*Generative artificial intelligence.* A large language model (Claude, Anthropic) assisted with drafting and editing the manuscript text and with writing the analysis and figure code, all under author review. It was not used to generate, annotate or interpret imaging data; every number reported here is computed by the released code from the released volumes.

### II.F. Validation

Validation has three layers. Geometry is checked first, because a measurement taken on a corpus with a geometric fault still returns a plausible-looking number.

*Geometric invariants.* Every case is checked for label geometry agreeing with its CT in shape, affine and voxel spacing; for identifiers within the published scheme; for a non-empty label; and for sided structures falling on the correct sides, with the left–right axis read from the affine rather than assumed.

The check compares *each sided pair separately.* Testing the ribs alone passed all 802 records, four of which carried a `left_hip` label on the patient's right side; pooling every sided structure into one test still missed one of those four records, because a large correctly-sided structure masks a smaller transposed one. A gate that passes everything is not evidence of a clean corpus until it has been shown capable of failing.

*Rib–vertebra incidence.* Each rib is matched to its nearest vertebra and compared against the vertebra its number implies. Of 11,548 ribs, 5,788 are not evaluable because the expected vertebra lies outside the field of view and 11 have no vertebra within the anchor distance; of the 5,749 evaluable, 5,747 match, two (0.035%) are offset by one level, and no case is misnumbered. The denominator is stated because quoting the two offsets against all 11,548 ribs would halve the rate by counting ribs the check never examined. Both offsets are field-of-view truncations, one at +1 level and one at −1.

The same check carries an invariant that reads as a null result. Of the 5,749 evaluable ribs, *zero* are assigned to a lumbar vertebra, not because no lumbar ribs exist, since 16 records carry one, but because such a rib takes class 60 or 61 and never enters the numbered set the test examines. A numbered rib landing on a lumbar vertebra would be a counting error of exactly the kind this dataset exists to expose.

*Agreement with published values.* Pelvic incidence is the strongest of these checks because it is a morphological property rather than a posture; it agrees to within half a degree with the automated supine CT series of Veilleux

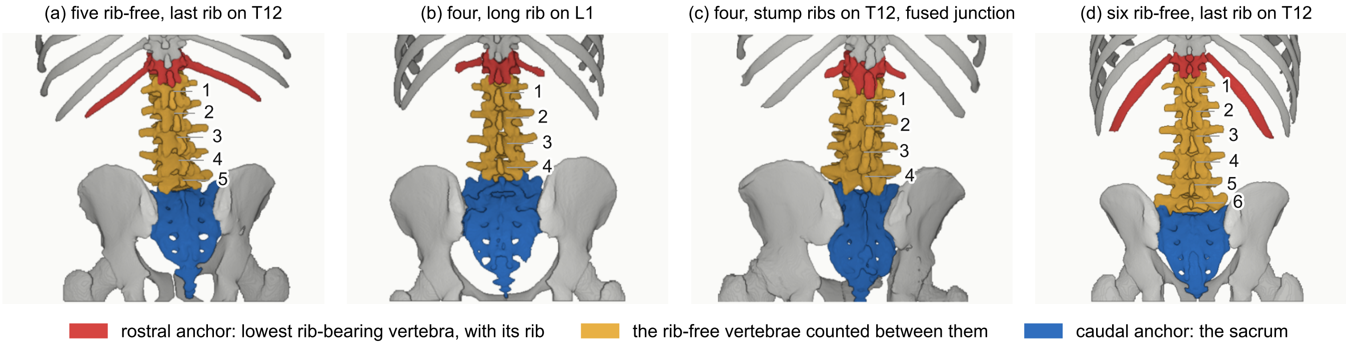


FIG. 2 The two anchors, from the released labels, aligned on the sacrum. Red, lowest rib-bearing vertebra and its rib; blue, the sacrum; yellow, the rib-free vertebrae between. Cases 0704, 0428, 0094 and 0005, chosen by rule as the most ordinary member of each phenotype (`scripts/select_anchor_cases.py`). (b) and (c) both count four. In (b) a long rib sits on L1; in (c) the twelfth ribs are stumps and the segment below the fourth lumbar body is fused to the sacrum (Castellvi IIIb).

*et al.*[25] and sits below the standing figure, the direction reported for subjects imaged both ways.[24] Sacral slope and pelvic tilt are postural, and read three degrees below and four above Veilleux's asymptomatic subjects; a second supine CT series reports 53.4, 34.1 and 19.2°, within a degree of this cohort on all three.[23] Recumbency runs the wrong way to explain the gap, raising slope rather than lowering it, so the residual tracks cohort and not construction.

### II.G. Castellvi typing

Every record whose vertebral count departs from five carries a Castellvi grade[39] in the released manifest, 33 cases typed I–IV with the $a/b$ unilateral–bilateral qualifier. The grade and the count are *different axes*. Grade IIIb occurs here at rib-free counts of four, five and six alike, and seven of the 33 graded cases carry a normal count of five.

Two radiology resident physicians graded every case independently and resolved their six disagreements by consensus; both reads and the consensus are in the manifest. The distinction the readers disagreed on, type II (articulation) against type III (bony fusion)[5], is the one that matters clinically, and four of the six disagreements were III against II.

### II.H. Surgical hardware, and why a threshold is not enough

Metal is trivial to detect and hard to interpret, and the difference matters here because **an iatrogenic fusion is indistinguishable from a congenital one to a distance measurement**. A cage-bridged interspace reads as "no gap" exactly as a fused transitional vertebra does.

*II.H.0.a. Detection over-calls by a factor of five.* A 2500 HU threshold inside a shell around the labeled skeleton flags 52 of the 802 records. All 52 were reviewed manually, confirming 11 and rejecting 41 (79%) as contrast, calcification and reconstruction artefact. Attenuation does not separate the two groups (both saturate); site and volume do, every confirmed implant at 2,586 $\text{mm}^3$ or above and every rejection at 1,768 $\text{mm}^3$ or below.

*II.H.0.b. The subtype block had to be extended.* Identifiers 62–65 name spinal instrumentation (generic, cage, screw-and-rod, plate) and cannot describe what this cohort holds, so **66 arthroplasty**, **67 sacroiliac screw** and **68 osteosynthesis** were added. Osteosynthesis holds parts of one bone together where arthroplasty replaces a joint, and a shape rule calls a femoral stem a rod. The confirmed cases are eight arthroplasties, one osteosynthesis, one sacroiliac fixation and one pair of interbody cages (Fig. 3).

*II.H.0.c. Metal outranks bone, and that is a subtraction.* In every confirmed case the implant was already inside a bone label, so naming it reclaims 1,538,852 voxels rather than adding them. In eight of these cases the femoral head that pelvic incidence and tilt are measured from is an implant.

## III. DATA FORMAT AND USAGE NOTES

Each record is a gzipped NIfTI image/label pair sharing an identical $4 \times 4$ affine, canonicalized to PIR, requiring no resampling. Masks are NIfTI rather than DICOM, which is what volumetric segmentation tooling expects.

Splits are patient-grouped and LSTV-stratified five-fold, frozen and shipped with the data as `splits_5fold.json`. The five validation folds partition all 802 records, so the release provides cross-validation and *not* a held-out test set; users reporting a single held-out number should carve one out and state which records it holds.

Stratification is on the transitional subtype rather than a binary LSTV flag. Across all 802 records the source transitional label reads lumbarization in 14, sacralization in 17 and semi-sacralization in two; separately, 18 records carry a sixth lumbar-type vertebra, nine carry only four lumbar labels, and 33 carry a consensus Castellvi grade. A record called sacralization by its four lumbar labels enters the same stratum as one called sacralization by the source label, so the two rare strata hold 15 records each and every validation fold carries three of each. That is the most stratification can guarantee, and it is why a fold-level metric on the rare classes carries an error bar far wider than its own decimal places.

TABLE II Spinopelvic measures, mean ± SD over the 756 of 802 records passing the geometric gates, against two supine CT series that report means.[23,25] Standing values are 55.0, 41.0, 13.0 and 60.0°.[38] The end-plate is fitted to the whole sacrum; the gate rejects 14 records where the previous carved-S1 fit lost 239. Pelvic incidence and tilt are right-skewed (0.25 and 0.33), so their medians—51.5 and 15.0°—sit below these means and the modal peak in Fig. 4 sits below both.

| Measure | This dataset (supine, $n = 756$) | Veilleux ($n = 200$) | Hasegawa ($n = 24$) |
|---|---|---|---|
| Pelvic incidence | 52.1 ± 11.1° | 52.1 | 53.4 |
| Sacral slope | 36.6 ± 10.6° | 36.5 | 34.1 |
| Pelvic tilt | 15.5 ± 7.9° | 15.6 | 19.2 |
| Lumbar lordosis | 51.7 ± 13.3° | — | — |
| PI − LL mismatch | 0.3 ± 10.3° | — | — |

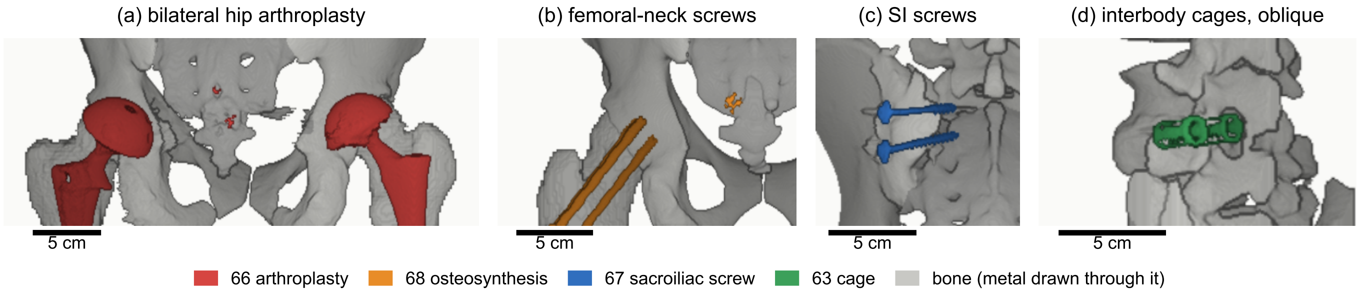


FIG. 3 Instrumentation in the release, one exemplar each, drawn through bone. (a) Hip arthroplasty, 66, $n = 8$. (b) Femoral-neck screws, 68, $n = 1$. (c) Sacroiliac screws, 67, $n = 1$. (d) Interbody cages, 63, $n = 1$. Bar, 5 cm.

TABLE III Data types and metadata, and the records populating each, from the released manifest.

| Field | Records | Available |
|---|---|---|
| Identifiers and paths, configuration, match type, image/label alignment check | 802 | 100.0% |
| Patient position, tube potential, section thickness, kernel | 802 | 100.0% |
| Transitional label and class, annotation origin, instrumentation flags | 802 | 100.0% |
| Spine series identifier; bone-fraction check | 782 | 97.5% |
| Scanner manufacturer and model | 768 | 95.8% |
| Sex | 749 | 93.4% |
| Age | 709 | 88.4% |
| Pelvic series identifier | 713 | 88.9% |
| Castellvi grade (consensus of two readers) | 33 | 4.1% |

The archive of record is Zenodo (https://doi.org/10.5281/zenodo.22139642). Code and documentation are on GitHub at the tagged release commit; any model-hub copy is a convenience mirror, not the archive.

## IV. DESCRIPTIVE ANALYSIS

The cohort is a colorectal cancer screening population of 802 records: 393 recorded female and 345 male, 11 carrying DICOM's *other* value and 53 no sex field, with age present for 709 (median 59, range 50–89). All 802 are counted in totals; the eleven outside the female/male dichotomy are excluded from sex-stratified measures.

*Count-free measures.* Five rib-free vertebrae above the sacrum is typical; four and six are where transitional anatomy sits, and which is present does not by itself determine what to call it. The lowest-rib length ratio is bimodal, and its lower mode, below 0.33 of the rib above, marks a stump twelfth rib in 98 of the 789 records with a measurable pair (12.4%, against 12.6% on whole-spine CT[9]); 16 records carry a lumbar rib (2.0%, against 1.5%). The co-occurrence reported there is testable here only where the junction was read, and only within those 789: they hold 14 of the 17 source-labeled sacralizations, and stump ribs accompany four of those 14 and 94 of the other 775 (odds ratio 2.9, $p = 0.08$, Fisher's exact test). No lumbarization carries a lumbar rib.

*Spinopelvic measures* (Fig. 4). Pelvic incidence sits on the standing reference, as a morphological property should; slope and tilt sit below and above the published CT values, which Table II places against two series.

*Opportunistic measures.* Every scan carries a vertebral bone density measurement at no additional dose.[40] L1 trabecular attenuation is reported at the published standard site.

## V. POTENTIAL APPLICATIONS AND LIMITATIONS

The principal use is the *spinopelvic interface.* Spine and pelvis carry radiologist-sourced annotation on one coordinate frame in 802 records, making sacral morphology measurable against the lumbar column rather than in isolation, among them the sacral endplate and its slope, the alar corridors constraining S2-alar-iliac and iliac screw trajectories, and the L5–S1 geometry a lumbosacral fusion must cross. Pelvic incidence is derived from the segmented sacrum and femoral heads, so it is reproducible from the labels themselves.

The transitional layer serves that use rather than competing with it, since no trajectory can be planned across a junction whose segments cannot be named.

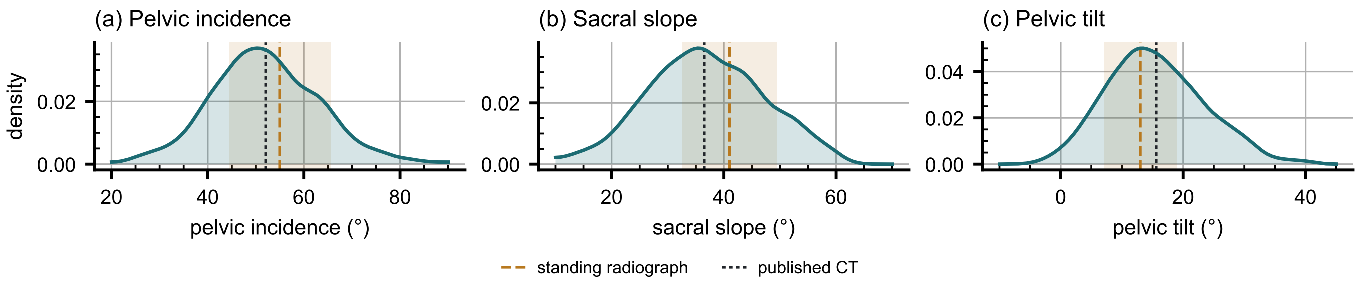


FIG. 4 Spinopelvic measures against standing reference bands (mean ± SD)[38].

*A proxy for the preoperative view.* No public preoperative lumbar CT cohort exists that we know of, so this release is the closest available stand-in for the view a lumbar case is planned on: it holds the T12-to-S1 span with the lowest ribs and the pelvis, 377 of its 802 records were acquired prone, the position of posterior lumbar surgery, and it carries classes for the anomalies that make level identification fail. It is a screening population at colonography dose, not a surgical series, so it supports method development for level identification and instrumentation geometry rather than validation of a planning decision.

### V.A. Three uses the released measurements support

*Naming a level from its own geometry.* Wrong-level spine surgery runs at roughly one in 3,110 spinal procedures, and among the predominant contributing factors is the transitional anatomy this cohort was assembled around.[41,42] Per-level body height, canal width, endplate width, transverse-process span and wedge ratio are released across all 802 records, and because the labels were assigned against the twelfth rib rather than by enumeration, a model trained on them is not learning the convention that fails.

*Reference morphometry with its spread.* The values a surgeon consults for endplate and canal dimensions and pedicle width come from cadaveric series of ten to thirty specimens[26–28] redrawn in the textbooks as one curve per level[44], and a standard error describes the mean, not the spread of individuals. The same measures are given here from 802 records with the distribution attached (Fig. 5), reproducing the classical pedicle widening into the lower lumbar spine. Both layers of that figure are population intervals rather than confidence intervals, and that distinction is what makes their widths comparable: a confidence interval bounds a parameter, a population interval bounds individuals,[43] and at $n = 12$ the two differ by $\sqrt{12}$.

*Patient-specific models.* Patient-specific finite-element modeling for alignment planning and implant design[45] needs spine, pelvis and femoral heads in one frame, which is this release's construction, and 351 patients give a within-patient postural change to check a prediction against.

*Limitations: coverage and cohort.* Thoracic ground truth is field-of-view limited (Fig. 6) and does not reach T1. Postural angles are supine; pelvic incidence is not postural, but it is modality-sensitive, reading below radiography for the same subjects.[24] End-plate and pedicle width are withheld at L5, where the transverse processes arise in front of the canal wall the body is cut at. Those two and the spinopelvic angles, which rest on the sacral end-plate, are the provisional part of this release; the planned revision takes body and sacral plate from a substructure label, and the version DOI carries it. The cohort is one protocol of a colorectal screening population aged 50 and over, so its distributions should not be read as a surgical one.

*Label strength varies by structure, and the release does not average over that.* Vertebral labels derive from radiologist-supervised source annotations, corrected where those sources were wrong; pelvic labels on records that lacked one are pseudolabeled. The rib layer is a pseudolabel whose review was triaged, so its guarantee is the absence of the failure modes that rule detects. Transitional labels come from two independent sources and are not uniformly adjudicated, which is why the measures reported here are count-free; the Castellvi grades are a consensus of two radiology resident physicians. The sacrum is the source annotation's own outer boundary throughout; no sub-division of it is asserted, the separate S1 label of earlier releases having been withdrawn as unreliable.

*Absent structures, and no classifier at the lower border.* Sacrum, hips and femora are present in all 802 records; one lacks T12, outside the field of view. Nine carry no L5 identifier because their source annotation counted four lumbar vertebrae; all nine are Castellvi IIIb, the fused segment is delineated with the sacrum as the caudal anchor, and the manifest names them. No shape-based classifier is attempted at the lumbosacral junction: 15 records in each rare stratum leave three per fold. The 0.990 area under the curve above is the thoracolumbar boundary, not this one.

## VI. DISCUSSION

*Comparison with related material.* VerSe[31,32] comes closest in intent, but carries no sacral mask for the structure a Castellvi grade describes, which this release segments as L6 or as sacrum, and neither the pelvis and femora a spinopelvic measurement needs nor prone acquisitions. What this release adds over TotalSegmentator[34] is the classes an anomaly needs, not more anatomy

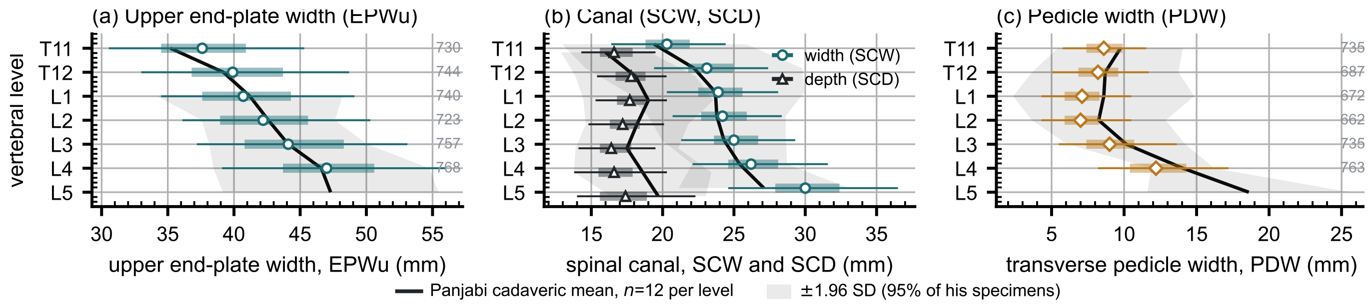


FIG. 5 Morphometry by level, T11–L5: median, interquartile range, 5th–95th percentile, $n$ at right. (a) End-plate width EPWu. (b) Canal width SCW and depth SCD. (c) Pedicle width PDW, both sides averaged. L5 is withheld in (a) and (c); see text. Reference is Panjabi's cadaveric series[26,27], keyed in (a), its SD recovered from the reported SEM; T11–T12 EPWu is digitised from his Figure 4A.

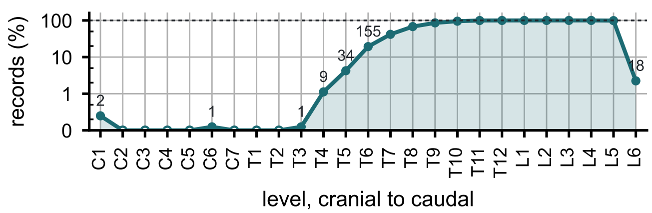


FIG. 6 Records carrying each vertebral level, cranial to caudal. Counts per identifier in Table S1.

per scan.

*Future directions.* Applying the released weights to VerSe would add the sacral anchor and the anomaly classes to a bone-kernel collection. A Castellvi read of the whole cohort would test the co-occurrence of rib and lumbosacral anomalies at full power and supply the labels a lumbosacral classifier needs. A corpus label from substructure segmentation[10] would measure the body without the canal cut that withholds L5 here.

## VII. CONCLUSION

CTSpinoPelvic1K places spine, pelvis, per-level ribs and femora on one coordinate frame in 802 records, and gives the anatomy that makes lumbar numbering ambiguous a class of its own. Because both counting anchors are explicit, a level can be identified from a field of view that cannot support the conventional count downward from C2.

## DATA AVAILABILITY

The release described here is v10, permanently archived at https://doi.org/10.5281/zenodo.22642578; the concept DOI https://doi.org/10.5281/zenodo.22139642 resolves to the current version. The archive carries the loader, the label scheme and the quality-control tables under an open-source licence. The repository additionally carries the extraction and figure code, the reference table with a citation on every row, the values plotted in Fig. 5, and a recipe for running them in order. Code: https://github.com/OpenSpineConsortium/CTSpinoPelvic1K (article materials under `paper/mpda/`); convenience mirror: https://huggingface.co/datasets/OpenSpineConsortium/CTSpinoPelvic1K; measurement toolkit: https://github.com/OpenSpineConsortium/OpenSpineToolkit.

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## CONFLICT OF INTEREST

The authors have no relevant conflicts of interest to disclose.

## ETHICS

This work uses publicly available, de-identified imaging and annotations derived from it. The authors' institutional review board determined that it is not human participant research under 45 CFR 46 and requires no oversight.

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